

Exercise 3.1.3 page 86 in Mining of Massive Datasets by J. Leskovec, A. Rajaraman and J. D. Ullman 3rd edition which states:

"Suppose we have a universal set U of n elements, and we choose two subsets S and T at random, each with m of the n elements. What is the expected value of the Jaccard similarity of S and T ?"

Solution

Let m be the number of elements in sets S and T that have been chosen randomly and independently from the set U containing n elements. Furthermore, elements in sets S and T are chosen from set U without replacement (i.e. each element in S and T can appear at most only once in respective set).

If sets S and T have k elements in common, then the Jaccard similarity of sets S and T will be equal to

$$\text{Jaccard similarity}(k - \text{common elements in } S \text{ and } T) = \frac{k}{2m - k}$$

The probability for sets S and T to have exactly k elements in common is equal to

$$\text{Probability}(k - \text{common}) = \frac{C_k^n C_{m-k}^{n-k} C_{m-k}^{n-m}}{C_m^n C_m^n}$$

Where C_k^n the binomial coefficient i.e. the number of ways to choose k elements out of n .

The above probability is justified as follows: C_k^n is the number of ways to choose the k common elements from the universal set U . Since each set S and T will have these elements as members (the k common), S will have an additional of $m-k$ elements that are only in S and T will have an additional of $m-k$ elements that are only in set T . The $m-k$ elements of S must be chosen from $n-k$ elements (since k are already in S) and the way to do this is C_{m-k}^{n-k} , while the $m-k$ elements of T must be chosen from $n-m$ elements (i.e. must be chosen from $n-k - (m-k) = n-m$ elements where the $m-k$ the number elements only in S) to make sure that no other common element will occur in S and T except for the k already chosen. The ways to do this is C_{m-k}^{n-m} . Applying the fundamental principle of counting the ways to choose sets S and T so that they have exactly k elements in common is $C_k^n C_{m-k}^{n-k} C_{m-k}^{n-m}$. The probability of choosing sets S and T so that they have exactly k elements in common is hence

$$Probability(k - common) = \frac{C_k^n C_{m-k}^{n-k} C_{m-k}^{n-m}}{C_m^n C_m^n} [1]$$

i.e. the number of ways for S and T to have exactly k elements in common divided by total number of ways to choose independently and without replacements the m elements of S and the m elements of T (without any constraint related to common elements) which is equal to $C_m^n C_m^n$.

The expected value of the Jaccard similarity is equal to the Jaccard similarity of having k elements in common multiplied by the probability for S and T to have k elements in common summing these products up over all possible values of k, i.e. from 0 to m. Hence, the expected value of the Jaccard similarity of sets S and T is given by the formula:

$$E(Jaccard\ similarity) = \sum_{k=0}^{m_1} \frac{k}{2m - k} \frac{C_k^n C_{m-k}^{n-k} C_{m-k}^{n-m_1}}{C_m^n C_m^n} \quad QED \quad [2]$$

Generalizing the formula

Below the generalization of formula [2] for the case where the sets S and T do not have the same number of elements. The reasoning is exactly the same as in the case where sets S and T have the same number of elements.

Let m_1 be the number of elements in set S and m_2 the number of elements in set T that have been chosen randomly from the set U containing n elements. Without harm we can assume that $m_1 \leq m_2$. Furthermore, elements in sets S and T are chosen from set U without replacement (i.e. each element in S and T can appear at most only once in respective set).

If sets S and T have k elements in common, then the Jaccard similarity of sets S and T will be equal to

$$Jaccard\ similarity(k - common\ elements\ in\ S\ and\ T) = \frac{k}{m_1 + m_2 - k}$$

The probability for sets S and T to have exactly k elements in common is equal to

$$Probability(k - common) = \frac{C_k^n C_{m_1-k}^{n-k} C_{m_2-k}^{n-m_1}}{C_{m_1}^n C_{m_2}^n}$$

Where C_k^n the binomial coefficient i.e. the number of ways to choose k elements out of n.

The above probability is justified as follows: C_k^n is the number of ways to choose the k common elements from the universal set k. Since each set S and T will have these elements as members (the k common), S will have an additional of m_1-k elements that are only in S and T will have an additional of m_2-k elements that are only in set T. The m_1-k elements of S must be chosen from $n-k$ elements (since k are already in S) and the way to do this is $C_{m_1-k}^{n-k}$, while the m_2-k elements of T must be chosen from $n-m_1$ elements (since $m_1 < m_2$) to make sure that no other common element will occur in S and T except for the k already chosen. The ways to do this is $C_{m_2-k}^{n-m_1}$. Applying the fundamental principle of counting the ways to choose sets S and T so that they have exactly k elements in common is $C_k^n C_{m_1-k}^{n-k} C_{m_2-k}^{n-m_1}$.

The probability of choosing sets S and T so that they have exactly k elements in common is

$$Probability(k - common) = \frac{C_k^n C_{m_1-k}^{n-k} C_{m_2-k}^{n-m_1}}{C_{m_1}^n C_{m_2}^n} [3]$$

i.e. the number of ways for S and T to have exactly k elements in common divided by total number of ways to choose the m_1 elements of S and the m_2 elements of T, without any constraint which is equal to $C_{m_1}^n C_{m_2}^n$.

The expected value of the Jaccard similarity is equal to the Jaccard similarity of having k elements in common multiplied by the probability for S and T to have k elements in common summing these products up over all possible values of k, i.e. from 0 to m_1 (since $m_1 \leq m_2$).

Hence, the expected value of the Jaccard similarity of sets S and T is given by the formula:

$$E(Jaccard\ similarity) = \sum_{k=0}^{m_1} \frac{k}{m_1 + m_2 - k} \frac{C_k^n C_{m_1-k}^{n-k} C_{m_2-k}^{n-m_1}}{C_{m_1}^n C_{m_2}^n} \quad QED \quad [4]$$

Lifting the constraint that $m_1 \leq m_2$ and generalize the result, in formula [4] the following replacements should be done:

$$m_1 = \min(m_1, m_2)$$

$$m_2 = \max(m_1, m_2)$$

and the formula becomes:

$$E(\text{Jaccard similarity}) = \sum_{k=0}^{\min(m_1, m_2)} \frac{k}{m_1 + m_2 - k} \frac{C_k^n C_{\min(m_1, m_2) - k}^{n-k} C_{\max(m_1, m_2) - k}^{n - \min(m_1, m_2)}}{C_{\min(m_1, m_2)}^n C_{\max(m_1, m_2)}^n} \quad QED \quad [5]$$

NOTE: The script implements the generalized formula [5].